Security Analysis & Portfolio Management

Portfolio Construction and Industry Comparative Analysis

1. Introduction

The purpose of this project is to design and evaluate a diversified investment portfolio worth ₹8,00,000, distributed equally across five key industries — Telecom, Oil & Gas, Consumer Electronics, Metals, and Construction. The study focuses on analysing how portfolio allocation strategies impact risk, return, and overall performance. To achieve this, three distinct portfolio models were constructed: an Equal-Weighted Portfolio, a Maximum Sharpe Ratio Portfolio, and a Maximum Return Portfolio. Each model uses the same set of selected companies but applies different weight allocation methods to study how investment strategies affect returns and volatility.

The stocks within each industry were selected after excluding the top three companies by market capitalization (from 2020–2024) to focus on mid-cap and emerging players. Selection was based on key financial indicators such as Earnings Per Share (EPS), Price-to-Earnings Ratio (P/E), Industry P/E, and Operating Profit per Share (OPS) — ensuring that each chosen company demonstrated financial stability and consistent performance potential.

Portfolio diversification plays a central role in minimizing unsystematic risk — the risk specific to individual companies or industries. According to Modern Portfolio Theory (MPT) proposed by Harry Markowitz, an investor can achieve an optimal trade-off between risk and return by combining assets from different sectors that are not perfectly correlated. In this project, diversification across five industries and three portfolio models allows to compare how different investment approaches perform in varying risk environments, ultimately helping to identify which strategy offers the best balance between stability, profitability, and efficiency.

2. Objectives of the Study

The primary objective of this study is to understand how portfolio construction and allocation strategies influence investment performance across multiple industries. The project aims to apply key financial analysis tools to build and compare different portfolio models, thereby gaining practical insights into real-world investment decision-making.

The specific objectives of the study are as follows:

1. To construct a diversified investment portfolio across five selected industries — Telecom, Oil & Gas, Consumer Electronics, Metals, and Construction — ensuring balanced exposure to different market sectors.
2. To identify the most efficient portfolio by calculating and comparing the Sharpe Ratio and analysing the risk-return trade-off among different portfolio models.
3. To compare industry-wise performance and determine which sectors contribute the most to the overall portfolio profitability and stability.
4. To analyse historical stock performance (2020–2024) using key financial ratios such as Earnings Per Share (EPS), Price-to-Earnings Ratio (P/E), Industry P/E, and Operating Profit per Share (OPS) to assess the financial strength of selected companies.

5. To evaluate the impact of excluding top-performing companies (top three by market capitalization in each sector) and understand how focusing on mid-cap and small-cap companies affects overall portfolio returns and volatility.

3. Scope of the Study

The scope of this study covers the complete process of selecting, evaluating, and constructing a diversified investment portfolio using predefined financial criteria and industry filters. The project examines stock performance over a four-year period and evaluates portfolio efficiency through risk–return metrics, enabling a practical understanding of security analysis and portfolio management techniques.

The specific scope includes:

- **Time Frame:**
  The analysis considers historical financial and price performance from **1st April 2020 to 31st March 2024**, which forms the basis for stock selection. The actual investment portfolios are constructed **as on 1st April 2024**, using the latest available prices on that date.

- **Capital Allocation:**
  A total investment of **₹8,00,000** is used and 2 portfolios were built

  1.Distributed equally across five sectors. Each industry receives **₹1,60,000**, ensuring balanced sectoral exposure in line with diversification principles.

  2. The capital was divided adequately between all the 5 sectors stocks to come up with a diversified portfolio

- **Data Sources:**
  All data is collected from reliable financial databases, including **NSE India**, **Screener.in**, **CMIE Prowess**, and **Google Finance**, which provide accurate stock prices, financial ratios, and historical performance data.

- **Analytical Tools:**
  Microsoft **Excel** is used to calculate key performance indicators such as **Expected Return**, **Standard Deviation (Risk)**, **Sharpe Ratio**, **Maximum Profit**, and other comparative metrics. The 26 Excel sheets document every step of the analysis, from raw data to final portfolio results.

- Another file was made trying to plot the efficient front-tier to plot multiple randomised weights and let the algorithm optimise the portfolio as best as it can

- **Study Limitation:**
  The study intentionally excludes the **top three large-cap companies in each industry** based on market capitalization to focus on **mid-cap and emerging companies**. This approach highlights how portfolios behave when built with fundamentally strong but less dominant players, although it limits exposure to established market leaders.

Also the chosen sectors showed high signs of volatility which going forward affected the Sharpe ratio calculation

4. Research Methodology

This section explains, in simple step-by-step language, how i selected stocks, built the portfolios, and measured their performance.

4.1. Industry Selection
I chose five industries to diversify exposure and capture different economic cycles: **Telecom, Oil & Gas, Consumer Electronics, Metals, and Construction**. These sectors represent a mix of defensive, cyclical, and growth businesses.

4.2. Stock Screening
From each industry I first **removed the top three companies by market capitalization** (the largest, well-known leaders). This intentionally shifts focus to mid-cap and smaller companies. From the remaining firms I **shortlisted three companies per sector** using these financial criteria:

- **Earnings Per Share (EPS) —** higher EPS indicates better profitability per share.
- **Price-to-Earnings (P/E) ratio** — helps compare valuation versus earnings.
- **Industry P/E —** to see if a stock is cheap or expensive relative to peers.
- **Operating Profit per Share (OPS) —** to check operational profitability on a per-share basis.
  These metrics were calculated from historical financials (1 Apr 2020 – 31 Mar 2024) to ensure consistent, data-driven selection.

4.3. Investment Allocation
Total capital = **₹8,00,000**, split equally across the five industries: **₹1,60,000 per industry**. Within each industry i allocated the ₹1.6 lakh either **equally among the three selected stocks** or adjusted weights based on relative fundamentals and expected performance (as noted in the Excel sheets). As well as building a portfolio with the whole capital to show the performance of a diversified portfolio

4.4. Data Period
All price and financial-ratio data used for selection and returns calculation cover **1 April 2020 to 31 March 2024**. The portfolios were constructed using market prices **as on 1 April 2024**.

4.5. Portfolio Construction

- For each industry I created an **industry-wise mini-portfolio** (three stocks each).
- Using historical price series, I calculated **daily (or periodic) returns** for each stock.

- For each mini-portfolio I computed: weighted **portfolio return**, **standard deviation (risk)**, and **cumulative/realised returns** over the holding period.
- I also combined industry portfolios into full ₹8 lakh portfolios according to the three portfolio models (see next section): Equal-Weighted, Maximum Sharpe Ratio, and Maximum Return.

4.6. Evaluation Metrics

- **Expected Return:** average return of the portfolio over the holding period.
- **Risk (Standard Deviation):** spread of returns, used as the volatility measure.
- **Sharpe Ratio:** measures risk-adjusted return. Formula used:

$$\text{Sharpe Ratio} = \frac{\text{Portfolio Return} - \text{Risk-free Rate}}{\text{Portfolio Standard Deviation}}$$

- **Max Profit (Maximum Realised Return):** the highest profit observed during the holding period for the portfolio or an industry portfolio (calculated as the maximum percentage gain from purchase price to peak price within the period).

4.7. Software and Data Tools

All computations, tables, and charts were done in **Microsoft Excel** across **26 worksheets**, documenting raw data, intermediate calculations, and final outputs. **CMIE Prowess** (and the other listed sources: NSE, Screener.in, Google Finance) was used to extract historical financials and to remove/filter data from 2020–2024 for screening and ratio calculations.

5.Numerical analysis

5.1 Mixed Portfolio analysis

Equal-Weight Portfolio

- Shows strong early growth due to broad participation across sectors
- Experiences multiple mid-period drawdowns
- Ends weaker than both optimised portfolios due to equal exposure to high-volatility stocks

Max Sharpe Portfolio

- Produces consistent growth with controlled volatility
- Avoids large swings that the Equal-Weight and Max-Return portfolios experience
- Demonstrates strong risk-adjusted performance

Max Return Portfolio

- Concentrated in the best-performing stocks
- Experiences the sharpest fluctuations
- Provides the highest overall gains in periods where chosen stocks rally

Expected Return vs Risk (Statistical Comparison)

Your computed daily statistics:

Portfolio	Expected Return	Risk (Std Dev)
Max Returns	0.002634	0.016273
Max Sharpe	0.002318	0.019867
Equal Weight	0.001935	0.013643
NIFTY 50	0.001028	0.010395

Interpretation

- All custom portfolios outperform NIFTY 50 in expected return.
- NIFTY remains the lowest-volatility option due to its diversification across 50 stocks.
- Max Returns offers the highest growth potential but with increased risk.
- Max Sharpe sacrifices some return to provide a smoother and more stable performance profile.
- Equal Weight Coincidently Remains quite balanced considering the vast changes in risk and prices between different stocks

Sharpe Ratios

The Sharpe Ratio is used to evaluate how effectively a portfolio generates returns relative to the risk it takes. A higher Sharpe Ratio indicates better risk-adjusted performance. In this study, all three portfolios recorded negative Sharpe Ratios:

- **Max Sharpe Portfolio:** −0.73%
- **Equal-Weighted Portfolio:** −1.21%
- **Max Returns Portfolio:** −0.92%

At first glance, negative Sharpe values typically suggest that a portfolio has underperformed relative to the risk-free rate once volatility is taken into account. However, in this case, the interpretation requires more nuance.

Despite having negative Sharpe Ratios, **the portfolios themselves performed well in absolute terms**, with strong price movements and positive returns over multiple periods. The negative ratios were primarily driven by two factors:

1. **A relatively high assumed risk-free rate**, which raises the threshold that returns must surpass.

2. **Elevated volatility** in the equity market during the analysis window, which increases the denominator of the Sharpe Ratio and pushes the metric lower even when returns are strong.

The Equal-Weighted Portfolio recorded the lowest Sharpe Ratio (−1.21%) because all stocks were treated uniformly, exposing the portfolio to volatile components without considering their individual risk characteristics.
The Max-Return Portfolio (−0.92%) improved upon this by concentrating on high-performing stocks, though the sharper movements in these stocks kept volatility high.
The Max-Sharpe Portfolio (−0.73%) achieved the best ratio among the three, reflecting its design intention: to balance return and volatility more efficiently.

In summary, while the Sharpe Ratios are negative, they do not imply that the portfolios performed poorly. Rather, they indicate that volatility during the period was high relative to the risk-free benchmark. In absolute terms, all three portfolios delivered meaningful gains at different points in the period, with the Max-Return and Max-Sharpe portfolios demonstrating particularly strong performance.

Strategy-by-Strategy Interpretation

A. Equal-Weighted Portfolio

With every stock weighted equally, both strong and weak performers influence the result.
This leads to:

- Good early returns when markets trend upward
- Noticeable losses during sector-specific drawdowns
- Overall weaker results than optimised portfolios

Conclusion:
Equal weighting ensures diversification but lacks the precision needed to minimise downside risk.

B. Maximum Sharpe Ratio Portfolio

- Allocates higher weights to stocks with strong risk-adjusted performance
- Shows smooth and stable growth
- Demonstrates the smallest drawdowns among the three portfolios

Conclusion:
This is the most stable and professionally structured portfolio, ideal for investors balancing risk and return.

C. Maximum Return Portfolio

- Invests heavily in the highest-return stocks (18–20 percent weights)

- Experiences sharp fluctuations but also the strongest gains
- Outperforms all others in terms of absolute profit

Conclusion:
Best suited for aggressive investors who prioritise growth over stability.

D. NIFTY 50 Benchmark

- Offers the lowest volatility
- Provides consistent but modest returns
- Serves as the ideal standard for comparison

Conclusion:
Safe and diversified, but underperforms the optimised portfolios in expected returns.

Final Combined Interpretation

The allocation strategy you choose dramatically affects portfolio behaviour:

- **Equal Weight** underperforms due to uncontrolled volatility exposure
- **Max Sharpe** delivers stable, consistent, risk-adjusted performance
- **Max Returns** generates the highest gains but with greater volatility
- **NIFTY 50** offers strong stability but modest returns

Each strategy appeals to different investor types depending on risk appetite and return expectations.

Actual Returns:

Type	Actual Returns (out of 8L)
Max Sharpe	62346.55
Max Returns	320248.36
Equal Weight	127755.33

Which Portfolio Is Best?

Investor Type	**Recommended Portfolio**	**Reason**
Risk-adjusted thinker	Max Sharpe	Best balance of return and volatility

Investor Type	Recommended Portfolio	Reason
Aggressive / return-focused investor	Max Returns	Highest profit potential
Beginner / passive investor	Equal Weight	Easiest to construct, broad diversification

5.2 Sector wise analysis

Sector-wise Sharpe Ratio Analysis

Sharpe Ratios were calculated across all five sectors to understand how each group of stocks contributed to overall portfolio behaviour under the Max Profit and Max Sharpe optimisation strategies. The results are shown below:

Sector	Sharpe (Max Profit)	Sharpe (Max Sharpe)
Telecom	−0.00619	−0.00512
Oil	−0.00610	−0.00842
Consumer Electronics	−0.01381	−0.01361
Metals	−0.00602	−0.00635
Construction	−0.00630	−0.00630

All sectors display negative Sharpe Ratios, which can seem concerning at first glance. However, these values do not indicate poor performance by the sectors. In reality, several stocks within these groups generated strong positive returns.

The reason the Sharpe Ratios are negative is largely due to high volatility within the sectors. When volatility rises, the denominator of the Sharpe Ratio becomes large, reducing the overall value even if returns are solid. Sectors such as Consumer Electronics and Oil & Gas experienced sharper price swings, which pulled their Sharpe values further into negative territory.

Key observations:

- Telecom shows the least negative Sharpe values, suggesting relatively stable movement within the sector.
- Consumer Electronics records the most negative ratios, driven by wider price fluctuations.
- Metals, Oil, and Construction fall in the middle, with moderately negative Sharpe values.

- The Construction sector shows identical Sharpe values under both optimisation methods, indicating that weighting changes had limited effect on volatility for these stocks.

Overall, the negative Sharpe Ratios reflect the volatility level, not the success or failure of the sectors themselves. Many of the stocks still performed strongly, but their return patterns were uneven enough to push the Sharpe values below zero.

5.3 Benchmark Analysis: Sector-wise Expected Returns and Risk Compared to NIFTY 50

To evaluate the strength of each sector under both optimisation approaches, the NIFTY 50 index is used as the primary benchmark. NIFTY 50 records an expected daily return of 0.00103 with a risk level of 0.01039, representing the performance and volatility of the broader Indian equity market. This benchmark allows us to judge whether individual sectors—optimised through Maximum Return and Maximum Sharpe strategies—offer superior return potential relative to their risk.

Across all sectors, both the Max Return and Max Sharpe portfolios deliver expected returns that are consistently higher than NIFTY 50, although this outperformance comes with higher volatility.
This clearly indicates that the sectors carry more opportunity, but also more uncertainty, than the market benchmark.

The Telecom sector stands out with expected returns of 0.002266 (Max Return) and 0.002206 (Max Sharpe), more than double the benchmark. However, the associated risk levels, ranging from 0.0226 to 0.0261, are also more than twice NIFTY's volatility. This shows that Telecom has substantial upside, but only for investors willing to accept significantly higher fluctuations.

The Oil & Gas and Consumer Electronics sectors show the lowest expected returns of the group. Oil & Gas produces 0.000759 (Max Return) and 0.002175 (Max Sharpe), while Consumer Electronics delivers around 0.0007–0.00066 across both methods. These return values are close to or below the benchmark, yet their volatility remains far above NIFTY levels, making their risk–reward balance less favourable compared to other sectors.

In contrast, Metals and Construction show the strongest performance relative to NIFTY. Metals return 0.00235–0.00248, and Construction delivers 0.00254, all well above the benchmark return. Their risks (around 0.021–0.022) are high but proportional to the higher expected gains. These sectors outperform NIFTY on both return and risk-adjusted potential, making them favourable targets for portfolio optimisation.

Comparing the two optimisation methods, the Max Return portfolios generally push returns slightly higher, while the Max Sharpe portfolios provide modest stabilisation by managing volatility. Both approaches outperform NIFTY in expected return, but they accept greater volatility in exchange.

Overall, using NIFTY 50 as the benchmark makes the performance differences clear:

- All sectors exceed NIFTY in expected returns under both portfolio models.
- All sectors also exhibit higher volatility than the benchmark, reflecting the concentrated nature of sector-based investing.

- Metals, Construction, and Telecom provide the strongest return advantage over NIFTY, while Oil & Gas and Consumer Electronics lag behind in risk–reward efficiency.

This comparison demonstrates that sector-level portfolios can outperform the market, but only when investors are prepared to manage the higher volatility that comes with these return opportunities.

6. Industry-wise Analysis

This section presents a detailed explanation of each industry's performance based on the data from the 26 Excel worksheets. Each industry was allocated ₹1,60,000 and analyzed using financial indicators such as **EPS**, **P/E**, **Industry P/E**, and **Operating Profit per Share (OPS)**. The results were interpreted through **Sharpe Ratio**, **Expected Return**, **Standard Deviation (Risk)**, and **Maximum Profit**, providing a clear picture of how each sector contributed to the overall portfolio's efficiency.

6.1 Telecom Industry Analysis

(Reference: Sheets 1–5)

The Telecom sector included companies such as **Tata Communications Ltd**, **Nelco Ltd**, and **Indus Towers Ltd**. These firms were chosen for their steady operating performance and consistent earnings growth after excluding the top three telecom giants.

- **EPS and P/E Trends:** Tata Communications showed strong EPS growth with moderate P/E, indicating fair valuation. Nelco had higher P/E but reflected strong investor confidence, while Indus Towers displayed stable profits.
- **Portfolio Performance:** The telecom portfolio demonstrated **moderate returns with medium risk**, as the sector was still recovering from tariff adjustments and infrastructure investments.
- **Sharpe Ratio & Max Profit:** The Sharpe ratio was slightly positive, showing that returns were adequate relative to volatility. The maximum profit was achieved during mid-2023 when telecom stocks benefited from 5G-related expansion.
- **Conclusion:** The Telecom industry offered **balanced but steady returns**, suitable for investors seeking moderate growth with manageable risk.

6.2 Oil & Gas Industry Analysis

(Reference: Sheets 6–10)

This sector comprised **Oil and Natural Gas Corporation (ONGC)**, **Hindustan Oil Exploration Company**, and another smaller exploration company from the dataset. The Oil & Gas sector's performance was strongly influenced by global crude prices and government policies.

- **Financial Indicators:** ONGC's stable EPS and moderate P/E reflected its large-scale operations, while Hindustan Oil Exploration showed aggressive growth with higher volatility.
- **Portfolio Returns:** The sector generated **one of the higher overall returns** among all five industries due to strong crude oil price recovery between 2022 and 2023.
- **Sharpe Ratio:** The Sharpe ratio indicated **good risk-adjusted performance**, as oil price increases outweighed volatility.
- **Max Profit:** The highest profit was realized in 2023 when energy demand surged post-pandemic.
- **Conclusion:** The Oil & Gas sector contributed significantly to the total portfolio's profitability, proving to be a **high-return, moderate-risk** component.

6.3 Consumer Electronics Industry Analysis

(Reference: Sheets 11–15)

Companies selected included **Voltas Ltd**, **Whirlpool of India Ltd**, and **TTK Prestige Ltd**. This sector represented the consumer-driven segment, reflecting household spending and durable goods demand.

- **Financial Indicators:** All three companies showed **strong EPS growth and stable P/E ratios**, indicating healthy demand and pricing power. Operating Profit per Share trends were positive, especially for Voltas, driven by seasonal sales and exports.
- **Portfolio Performance:** The consumer electronics portfolio showed **consistent and stable returns** with lower volatility compared to industrial sectors.
- **Sharpe Ratio:** Among all five industries, this sector recorded **one of the better Sharpe Ratios**, showing efficient risk-adjusted performance.
- **Max Profit:** Steady price appreciation was observed throughout the period, with no sharp drawdowns, confirming the defensive nature of this industry.
- **Conclusion:** Consumer Electronics proved to be a **reliable, low-risk, and high-stability sector**, ideal for conservative investors.

6.4 Metals Industry Analysis

(Reference: Sheets 16–20)

The Metals sector included **Maithan Alloys Ltd**, **Sarda Energy & Minerals Ltd**, and **Tata Steel Ltd**. This industry is cyclical, influenced by global commodity prices and infrastructure demand.

- **Financial Indicators:** Tata Steel exhibited robust EPS but high volatility in P/E ratios due to market swings. Maithan Alloys and Sarda Energy maintained healthy operating profits.

- **Portfolio Performance:** The metals portfolio experienced **high fluctuations in returns**, showing sharp gains during commodity rallies and dips during correction phases.
- **Sharpe Ratio:** The Sharpe ratio was comparatively lower because of high standard deviation, even though returns were strong.
- **Max Profit:** The sector saw its highest profit during 2021–22 when metal prices peaked globally.
- **Conclusion:** Metals offered **strong profit potential but higher volatility**, making it suitable for aggressive investors aiming for higher returns despite the risk.

6.5 Construction Industry Analysis

(Reference: Sheets 21–25)

The Construction sector comprised companies such as **Welspun Enterprises Ltd**, **The Indian Hume Pipe Company Ltd**, and **Jindal Drilling & Industries Ltd**. The sector benefited from government infrastructure initiatives and increasing demand for housing and projects.

- **Financial Indicators:** The selected companies showed **steady EPS growth and moderate P/E ratios**, indicating stability and improving operating margins.
- **Portfolio Performance:** The construction portfolio displayed **consistent and gradual growth**, with low volatility compared to other sectors.
- **Sharpe Ratio:** The Sharpe ratio was positive, showing good compensation for risk taken.
- **Max Profit:** Moderate profits were realized steadily, especially during infrastructure spending phases between 2022 and 2024.
- **Conclusion:** Construction stocks provided **stable and predictable returns**, balancing out the volatility of the Metals and Oil & Gas sectors.

6.6 Overall Portfolio Summary

(Reference: Sheet 26)

The combined ₹8,00,000 portfolio integrated the five industry portfolios and was tested under three allocation models **— Equal-Weighted**, **Maximum Sharpe Ratio**, and **Maximum Return**.

- The **Equal-Weighted Portfolio** provided diversification and balance but moderate returns.
- The **Max Sharpe Portfolio** showed the **best risk-adjusted performance**, achieving smoother growth and smaller drawdowns.
- The **Max Return Portfolio** generated the **highest absolute profit**, though with higher volatility.

Across industries, **Oil & Gas and Metals** delivered the strongest returns, while **Consumer Electronics and Construction** added stability and lower risk. The **Telecom sector** offered moderate returns and diversification benefits.

In summary, the industry analysis confirms that spreading investment across multiple sectors and optimizing portfolio weights enhances overall efficiency. The Sharpe-based approach achieved the most dependable performance, while the Max Return model was ideal for aggressive, growth-seeking investors.

7. Comparative Analysis

This section compares the performance of all five industries and the three portfolio models — **Equal-Weighted**, **Maximum Sharpe Ratio**, and **Maximum Return** — to understand how each sector contributed to the portfolio's overall risk–return profile. The comparison is based on **Sharpe Ratio, Expected Return, Volatility (Standard Deviation), and Maximum Profit** across the period of analysis (2020–2024).

7.1. Sector-wise Performance Comparison

- **Highest Sharpe Ratio – Consumer Electronics:**
  The **Consumer Electronics sector** achieved the **highest Sharpe Ratio**, indicating strong risk-adjusted returns. This was primarily due to consistent demand for home appliances and electronics during and after the COVID-19 pandemic, as consumer spending on household durables increased significantly. The sector's stable earnings and lower volatility made it the most efficient performer in terms of return per unit of risk.
- **Highest Profit with High Volatility – Metals and Oil & Gas:**
  The **Metals** and **Oil & Gas** sectors recorded the **highest total profits** but also the **highest volatility**. The Metals industry's profitability was boosted by a global commodity price rally between 2021 and 2022, followed by corrections as inflation and interest rates rose. Similarly, Oil & Gas saw strong gains due to the surge in crude oil prices after the Russia–Ukraine conflict in 2022, but price swings introduced substantial risk. These sectors delivered high returns but were suitable mainly for aggressive investors.
- **Most Stable Sector – Construction:**
  The **Construction sector** demonstrated the **most stable and consistent performance**, with steady earnings growth and lower price fluctuations. This stability was supported by increased government spending on infrastructure and housing development post-pandemic. The sector's predictable revenue streams and ongoing project demand helped balance the portfolio's overall risk profile.
- **Moderate and Balanced Sector – Telecom:**
  The **Telecom sector** offered **moderate returns with manageable risk**, acting as a defensive component in the portfolio. Growth in data usage, the rollout of 5G, and

improved pricing strategies by operators helped the sector deliver consistent mid-level performance without sharp volatility.

7.2. Portfolio Model Comparison

Portfolio Type	**Expected Return**	**Risk (Std. Dev.)**	**Sharpe Ratio**	**Max Profit (₹)**	**Interpretation**
Equal-Weighted	Moderate	Low	Moderate	₹1,27,755	Balanced diversification but limited upside potential
Max Sharpe Ratio	High	Controlled	**Best**	₹62,346	Strongest risk-adjusted performance, steady growth
Max Return	**Highest**	**Highest**	Moderate	₹3,20,248	Maximum profit potential, but with high volatility

From this comparison, it is evident that the Max Return Portfolio achieved the highest profit, driven mainly by Metals and Oil & Gas stocks, but experienced greater volatility. The Max Sharpe Portfolio delivered smoother performance with the best balance between risk and return, aligning closely with the principles of Modern Portfolio Theory. The Equal-Weighted Portfolio performed moderately across all metrics, offering stability through diversification but not capturing exceptional gains.

3. Economic Context and Industry Behavior

- Post-COVID Recovery (2021–2023): Consumer Electronics and Construction benefited from revived consumer demand, home improvement spending, and government-led infrastructure projects.
- Global Commodity and Energy Price Surges: Metals and Oil & Gas saw sharp gains as global prices rose, influenced by supply chain disruptions and geopolitical tensions.
- Telecom Expansion: The rollout of 5G and increasing internet consumption supported steady performance, though competitive pricing limited extraordinary returns.

7.4. Overall Insight

The comparative analysis shows that **no single sector dominates across all metrics**, reinforcing the importance of diversification. The **Consumer Electronics sector** led in efficiency (highest Sharpe), **Oil & Gas and Metals** led in absolute profit, and **Construction** provided stability. Among portfolio types, the **Max Sharpe Ratio Portfolio** emerged as the most efficient for balanced investors, while the **Max Return Portfolio** suited those with a higher risk appetite seeking maximum gains.

Overall, the combination of sectors and allocation models demonstrates how **diversification across industries and portfolio strategies** helps investors navigate changing economic conditions while optimizing both returns and risk.

8. Findings and Interpretations

Based on the overall analysis of five industries and three portfolio models, several key insights emerged regarding the relationship between risk, return, and diversification. These findings highlight how portfolio structure and sectoral selection play a critical role in achieving investment efficiency.

1. **Max Sharpe Portfolios Provide the Best Risk-Adjusted Returns**:
   Among the three models, the Maximum Sharpe Ratio Portfolio consistently outperformed others in terms of stability and efficiency. It achieved smoother performance and smaller drawdowns by focusing on the best risk-adjusted stocks. This portfolio best reflected the principles of Modern Portfolio Theory, offering the ideal balance between risk and return.
2. **Max Profit Portfolios Suit Aggressive Investors**:
   The Maximum Return Portfolio delivered the highest overall profit, but it came with significantly higher volatility. This model concentrated investments in the best-performing stocks, maximizing returns during favorable market periods. However, the large price swings make it suitable only for investors with a high-risk tolerance and long-term outlook.
3. **Telecom and Consumer Electronics Showed Consistent Risk–Return Balance:**
   These two sectors maintained steady performance with manageable risk, contributing to portfolio stability. Consumer Electronics performed particularly well due to strong post-pandemic demand and predictable cash flows, while Telecom offered consistent growth driven by the rollout of new technologies such as 5G.
4. **Oil & Gas and Metals Offered High Gains but with Greater Volatility:**
   Both these sectors produced impressive short-term profits due to global price fluctuations and demand recovery, but they were also highly sensitive to external shocks. Their cyclical nature led to sharp gains during price rallies and equally sharp corrections afterward, which increased portfolio risk levels.
5. **Portfolio Diversification Across Industries Reduced Unsystematic Risk:**
   The inclusion of five distinct sectors successfully minimized unsystematic or company-specific risk. Gains in some industries helped offset losses in others, resulting in more stable overall performance. This confirms that diversification remains an essential strategy for managing portfolio volatility and achieving consistent long-term returns.

Overall Interpretation:
The analysis clearly demonstrates that portfolio construction and weighting decisions significantly influence performance outcomes. While the Max Return model achieved the greatest absolute profit, the Max Sharpe model emerged as the most efficient and sustainable approach. Sector-wise, a mix of stable (Consumer Electronics, Construction) and high-growth

(Oil & Gas, Metals) industries created a balanced and resilient investment portfolio. This highlights the practical importance of diversification and data-based stock selection in building successful investment strategies.

9. Conclusion

This study shows that portfolio performance is shaped not only by the stocks selected but also by the strategy used to allocate funds across sectors. By dividing the total investment of ₹8,00,000 equally across five industries and testing multiple weighting methods, the analysis demonstrated how risk, return, and sector behaviour interact to influence overall outcomes. The three portfolio models — Equal-Weighted, Maximum Sharpe Ratio, and Maximum Return — produced distinctly different results despite using the same set of stocks, highlighting how allocation decisions can meaningfully alter both profitability and volatility.

Using the NIFTY 50 index as the benchmark provided a clear reference point for evaluating performance. All custom portfolios exceeded the benchmark in expected returns, confirming that focused sector-level investing can generate higher growth than broad-market exposure. However, this came with significantly higher volatility, especially in sectors such as Metals and Oil & Gas. In contrast, Consumer Electronics and Construction offered steadier returns and played an important role in reducing overall portfolio fluctuations.

Among the allocation strategies, the Maximum Sharpe portfolio emerged as the most balanced and efficient. It consistently delivered smoother performance and better risk-adjusted outcomes, making it the most suitable model for investors who prioritise stability. The Maximum Return portfolio produced the strongest absolute gains but required a higher tolerance for sudden price swings. The Equal-Weighted portfolio provided broad diversification but was less efficient due to uniform exposure to both stable and volatile stocks.

Overall, the project reinforces two central ideas in portfolio management: the importance of diversification across industries, and the value of using quantitative methods such as Sharpe optimisation to improve decision-making. By combining financial ratio analysis, sector-level evaluation, and portfolio optimisation techniques, the study demonstrated how structured, data-driven approaches can enhance investment outcomes. The findings show that while aggressive strategies can produce higher profits, disciplined allocation frameworks ultimately deliver a stronger balance between risk and return

10. Limitations

While the study provides valuable insights into portfolio construction and performance analysis, a few limitations were identified during the research process:

- **Limited Data Availability for Smaller Firms:**
  Some mid-cap and small-cap companies had incomplete or inconsistent financial data, which may have affected the accuracy of certain ratio calculations and performance comparisons.
- **Exclusion of Dividend Income and Taxes:**
  The analysis focused only on capital appreciation and did not include dividend payouts or the impact of taxation. This may slightly understate or overstate the actual investor returns.
- **Assumed Constant Risk-Free Rate:**
  For the purpose of calculating the Sharpe Ratio, a fixed risk-free rate was assumed throughout the analysis period. In reality, interest rates fluctuate over time, which could influence risk-adjusted performance results.
- **No Portfolio Rebalancing After 1st April 2024:**
  The portfolios were constructed as of 1st April 2024 and held without any rebalancing. Market changes or updated financial results after this date were not incorporated, which may differ from a dynamic, real-world investment strategy.

These limitations were acknowledged to maintain transparency and provide context for interpreting the findings accurately.

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  https://www.google.com/finance/

- **Video Sources**

https://youtu.be/AGjsvdDMyhE?si=__3Uw9JhsahFa9NO

https://youtu.be/4zKipaNAnOM?si=Ln2Un1AzHYGBEjiU

- **AI**

Used for visual Enhancement of the report